

Ultra-Low Bitrate Perceptual Image Compression with Shallow Encoder (AEIC) on ImageNet-1k

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Multimedia Systems — project (paper: Zhang, Liu, Chen, CVPR 2026)

June 15, 2026

Paper: [arXiv:2512.12229](https://arxiv.org/abs/2512.12229) Code: github.com/LuizScarlet/AEIC

The problem

- Ultra-low bitrate compression: < 0.05 bits per pixel — for satellite/IoT/edge uplinks.
- Classic codecs collapse here (blur, blocking); *generative* codecs synthesize plausible detail at the decoder.
- But existing generative codecs put **huge pretrained encoders** (VAE/tokenizers) on the *sender* — exactly the weak device!
- Paper's question: **how shallow can the encoder be at ultra-low bitrates?**
Insight: less rate $\Rightarrow$ less information to extract $\Rightarrow$ much simpler encoder suffices.

The AEIC algorithm — architecture, training, distillation

- 1 **Shallow encoder** g_a : tiny StarNet CNN, image $\rightarrow$ latent ($32\times$ downsampling).
AEIC-ME: 3.09M params / 204 KMACs/px AEIC-SE: **0.94M / 46 KMACs/px, 35.8 FPS @1080p**
- 2 **Entropy model**: scale-mean hyperprior + quadtree spatial context;
Gaussian-mean rounding + rANS arithmetic coding.
- 3 **One-step diffusion decoder**: latent $\rightarrow$ SD-Turbo latent space $\rightarrow$ LoRA-adapted UNet, *single* denoising step (+ structural residual) $\rightarrow$ pruned VAE decoder. All generative “heavy lifting” happens at the receiver.

Training: relaxed-rate pretraining (MSE+LPIPS) $\rightarrow$ ultra-low-rate GAN fine-tune (DINOv3 discriminator, DISTS) $\rightarrow$ high-res fine-tune.

Dual-side distillation ME $\rightarrow$ SE (encoder latents + decoder/UNet features) shrinks the shallow encoder’s quality gap to $< 3\%$. Released: 5 rate points (ft2–ft32).

Our implementation (Google Colab, T4 GPU)

- Official repo + released AEIC-SE checkpoints; no retraining.
- **Data:** ImageNet-1k validation, streamed from Hugging Face; first 500 images with short side ≥ 256 px, exported losslessly as PNG.
- **5 small code changes:**
 - ① C++ rANS engine made optional (bpp from entropy-model likelihoods; verified vs. real bitstreams),
 - ② xformers $\rightarrow$ PyTorch SDPA attention,
 - ③ `torch.compile` off (ImageNet sizes vary $\Rightarrow$ constant recompilation),
 - ④ local port of `update_patch_fid` (FID/256) — `neuralcompression` no longer builds,
 - ⑤ KID subset size 1000 $\rightarrow$ 50 (small images = few 256px patches).
- Pinned `diffusers==0.25.1`, `transformers==4.46.3`, `peft==0.13.2`; kept Colab's PyTorch.

Metrics (same as the paper)

- **bpp** — rate: compressed bits / pixels.
- **PSNR, MS-SSIM** $\uparrow$ — pixel/structural fidelity (“distortion”).
- **LPIPS, DISTS** $\downarrow$ — deep-feature perceptual similarity to the original.
- **FID, KID** $\downarrow$ — distance between Inception feature *distributions* (patch-wise, MS-ILLM protocol) — realism without pixel alignment.

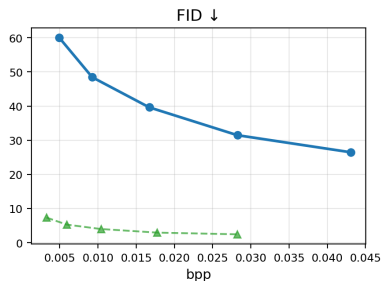
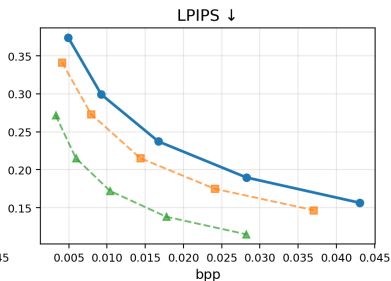
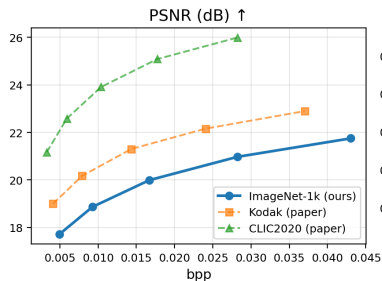
Together: the **rate–distortion–perception** trade-off. Generative codecs willingly trade PSNR for LPIPS/DISTS/FID at these rates.

Problems faced & solutions

- **Dependency conflicts** (repo pins torch 2.1 + old diffusers; neuralcompression fails to build) → pin only API-critical libs, keep Colab PyTorch, inline the one function used from neuralcompression.
- **Gated ImageNet** → HF token + streaming (no 6.7 GB download); ungated mirror as fallback.
- **C++ entropy coder build** (fetched pybind11 v2.10.4 predates Python 3.12) → one-line guard for the main pipeline; bumped pybind11 to v2.13.6 for the verification build: real 0.0180 vs. estimated 0.0167 bpp ($\approx 7\%$ header/rANS overhead).
- **Variable image sizes** → disable torch.compile; filter < 256 px (MS-SSIM constraint).
- **Colab session limits** → per-checkpoint logs/outputs; resumable runs.

Results on ImageNet-1k val (500 images)

ckpt	bpp	PSNR	MS-SSIM	LPIPS	DISTS	FID	KID
ft32	0.0049	17.71	0.576	0.374	0.202	60.1	0.0064
ft8	0.0167	19.99	0.728	0.237	0.144	39.6	0.0015
ft2	0.0431	21.75	0.825	0.157	0.106	26.5	-0.0002
JPEG _{q=1}	0.1915	21.07	0.809	0.547	0.411	151.5	0.0651



smaller ft number = more bits = better; full 5-point, 7-metric figure in the report

One image across the full rate sweep



- Degradation is gradual and content-dependent: printed text and fine patterns (top) are re-synthesized ever more loosely; the fur and thin wire fence (bottom) are visibly redrawn only at the lowest rate.
- Subjectively: down to ft16 (0.009 bpp) hard to distinguish from the original at normal viewing distance; at ft32 (0.005 bpp), differences are visible at any distance.

Interpretation

- All 7 metrics improve monotonically with rate ($0.0049 \rightarrow 0.0431$ bpp) — same trade-off shape as the paper's curves: **the codec generalizes** to unseen data.
- $\text{bpp} \approx 50\%$ higher than CLIC at the same checkpoint (0.0049 vs. 0.0033): small images (~ 0.2 MP) amortize the fixed hyper-latent + padding overhead poorly.
- MS-SSIM almost identical to Kodak (0.576–0.825 vs. 0.583–0.822), PSNR ~ 1 dB lower: at < 0.05 bpp the codec *hallucinates* texture rather than preserving pixels (and ImageNet “originals” are JPEGs themselves).
- FID 26.5–60.1 $\gg$ CLIC's 2.5–7.4 — only ~ 600 patches vs. $\sim 20\text{k}$ (FID is biased upward at small sample sizes); the *trend* is the signal. KID at 0.043 bpp is ≈ 0 (even slightly negative): reconstructions statistically indistinguishable.
- **JPEG can't reach this regime**: its floor is 0.19 bpp ($4.4\times$ ft2) with similar PSNR but LPIPS 0.55 vs. 0.16 and FID 151 vs. 26 — PSNR alone misleads.

Possible improvements & conclusion

Improvements

- Resolution-adaptive overhead (hyper-latent/padding) for small images.
- One variable-rate model instead of 5 checkpoints (rate conditioning).
- Distill the one-step UNet → lightweight *decoding* too; semantic bit allocation (faces/text); video extension.

Conclusion: AEIC runs on ImageNet-1k with only 5 minor code changes and keeps its rate–distortion–perception behaviour on unseen data — supporting the claim that at ultra-low bitrates a 0.94M-parameter encoder is enough when paired with a one-step diffusion decoder.